Course 4011 | Semester IV | Theory | 4 Credits

Design of Steel Structures

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Civil Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 4011 as Design of Steel Structures in Semester IV for Civil Engineering. The official SITTTR detailed source was unavailable during this cycle. With user approval, explanatory coverage uses a public diploma steel-structures curriculum and a public technical overview; it is not represented as recovered official SITTTR Course 4011 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders. All numerical design must follow the currently applicable Indian standards, project specifications and qualified engineering supervision; this handbook supports learning only and is not a construction design certificate.

Verified source note: The official SITTTR detailed source was inaccessible. Public source material supporting sections, connections, members, beams and trusses is linked in References.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Design of Steel Structures develops safe, code-aware reasoning for structural-steel members and connections. It connects material/section properties, loads and load paths to the behaviour and preliminary design of tension members, compression members, beams, trusses and their details.

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Prerequisite foundation

Engineering mechanics, strength of materials, basic structural analysis, dimensionally consistent calculation and drawing practice.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Identify structural-steel sections, properties, loads and basic design-limit concepts.
2. Explain bolted and welded connections and their role in safe load transfer.
3. Analyse the behaviour and preliminary design checks of tension, compression and flexural members.
4. Interpret simple steel-truss and structural drawings with fabrication, erection and safety awareness.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain structural-steel properties, section designation, loads and design assumptions.	Understand
CO2	Analyse the load transfer and basic checks of bolted and welded connections.	Apply
CO3	Apply preliminary checks to tension, compression and beam members using the applicable code/design data.	Apply
CO4	Interpret and communicate simple truss/member/detail drawings with safe fabrication and erection considerations.	Apply

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Steel, Sections, Loads and Design Basis	Properties of structural steel, common rolled/built-up sections, section notation, loads, limit states, stability, drawings and code awareness.	Approx. 14 hours	CO1	Module 1
2	Steel Connections	Bolted and welded connections, force transfer, detailing, eccentricity, fabrication, inspection and corrosion/fire considerations.	Approx. 16 hours	CO2	Module 2
3	Tension, Compression and Flexural Members	Tension capacity, net/gross section concepts, column buckling/effective length/slenderness, restrained beams, lateral stability and serviceability.	Approx. 16 hours	CO3	Module 3
4	Trusses, Drawings, Fabrication and Erection	Roof truss form, purlins, joint details, loads, simple member force interpretation, working drawings, quantity awareness, fabrication/erection safety and inspection.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Steel, Sections, Loads and Design Basis

Properties of structural steel, common rolled/built-up sections, section notation, loads, limit states, stability, drawings and code awareness.

CO1 · Approx. 14 hours

Steel Connections

Bolted and welded connections, force transfer, detailing, eccentricity, fabrication, inspection and corrosion/fire considerations.

CO2 · Approx. 16 hours

Tension, Compression and Flexural Members

Tension capacity, net/gross section concepts, column buckling/effective length/slenderness, restrained beams, lateral stability and serviceability.

CO3 · Approx. 16 hours

Trusses, Drawings, Fabrication and Erection

Roof truss form, purlins, joint details, loads, simple member force interpretation, working drawings, quantity awareness, fabrication/erection safety and inspection.

CO4 · Approx. 14 hours

MODULE 1

Steel, Sections, Loads and Design Basis

Properties of structural steel, common rolled/built-up sections, section notation, loads, limit states, stability, drawings and code awareness.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

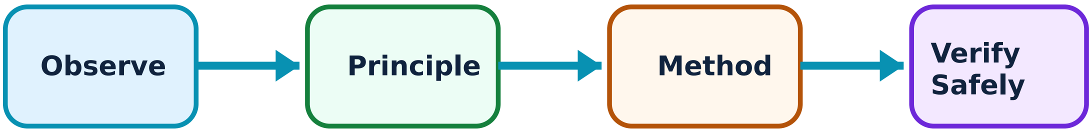
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Steel, Sections, Loads and Design Basis: identify the system, state its principle, follow the correct method and verify the result safely.

1. Structural steel and section forms−

Steel has high strength-to-weight ratio and ductility, but members must be selected for the force, connection, fabrication, corrosion and stability context. Common forms include angles, channels, I/H sections, tees, plates and hollow sections.

Study and application method

For a member, identify force type, likely failure/stability concern, feasible section family and the information that must be checked from a current standard/table.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For a member, identify force type, likely failure/stability concern, feasible section family and the information that must be checked from a current standard/table.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never choose a section from appearance or a catalogue size alone; verify grade, properties, slenderness, connection and load path.

2. Loads, load paths and limit states−

Dead, imposed, wind, seismic and other project loads travel through roof/floor systems, beams, columns, connections and foundations. Design considers safety and serviceability under specified combinations.

Study and application method

Sketch the structural system, draw force arrows, identify reactions and state the load combination/source that an authorised design would use.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

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Common mistake / precaution: Do not use isolated loads without applicable combinations, restraints and boundary conditions.

3. Standards, drawings and design records−

Design standards define material, load, detailing and verification requirements. Drawings communicate geometry, member marks, connections, tolerances and fabrication/erection information.

Study and application method

Read title block, grid, member schedule, section/detail callouts and notes; maintain units and a transparent calculation/drawing revision trail.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Only current project-approved codes and drawings may guide real construction.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Steel Connections

Bolted and welded connections, force transfer, detailing, eccentricity, fabrication, inspection and corrosion/fire considerations.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

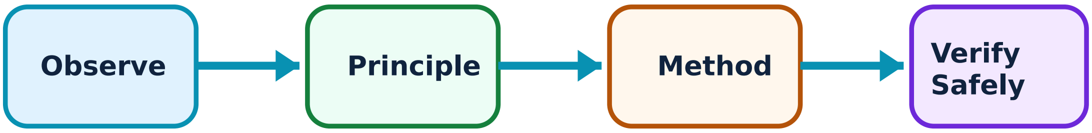
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Steel Connections: identify the system, state its principle, follow the correct method and verify the result safely.

1. Bolted connections−

Bolted connections transfer forces through bearing, shear, friction or tension mechanisms depending on type and detailing. Layout, edge distances, spacing and hole details affect strength and constructability.

Study and application method

Draw the force path from member to connection to support; list bolt type/grade, load direction, likely failure modes and code checks required.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw the force path from member to connection to support; list bolt type/grade, load direction, likely failure modes and code checks required.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not invent bolt capacities or spacing; obtain them from current approved design provisions.

2. Welded connections−

Butt and fillet welds create continuous load transfer when designed, fabricated and inspected correctly. Weld size, length, orientation, access and quality control are critical.

Study and application method

Show weld symbol/location, resolve applied force into the weld group and identify required design/inspection information.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Show weld symbol/location, resolve applied force into the weld group and identify required design/inspection information.

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Common mistake / precaution: Welding safety, qualified procedures and inspection are essential; classroom calculations do not authorise field welding.

3. Detailing and constructability−

A good connection is not only strong: it is accessible for fabrication, erection, tightening/welding, inspection, drainage and maintenance. Eccentric load paths and abrupt geometry can create unintended stresses.

Study and application method

Review a connection drawing for load path, access, sequencing, edge clearances, corrosion protection and inspection points.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Review a connection drawing for load path, access, sequencing, edge clearances, corrosion protection and inspection points.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never ignore erection stability or temporary bracing while focusing only on final connection strength.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Tension, Compression and Flexural Members

Tension capacity, net/gross section concepts, column buckling/effective length/slenderness, restrained beams, lateral stability and serviceability.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

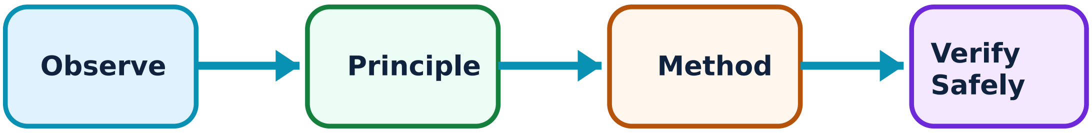
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Tension, Compression and Flexural Members: identify the system, state its principle, follow the correct method and verify the result safely.

1. Tension members−

Tension members such as ties and truss elements can fail by yielding, rupture at net section or connection-related mechanisms. Section reduction at holes and connection geometry matter.

Study and application method

Identify force, connection layout and critical section; distinguish gross/net area concepts and list relevant current-code strength checks.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify force, connection layout and critical section; distinguish gross/net area concepts and list relevant current-code strength checks.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not assume the full section area remains effective where bolts, weld geometry or eccentricity alter force flow.

2. Compression members and buckling−

Compression members may fail by material yielding or instability. Effective length, end restraint, slenderness, imperfections and local buckling influence capacity.

Study and application method

Sketch end conditions and likely buckling axis, determine the data needed for effective length/slenderness and explain why bracing changes behaviour.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch end conditions and likely buckling axis, determine the data needed for effective length/slenderness and explain why bracing changes behaviour.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A member with sufficient material strength can still buckle; never neglect stability checks.

3. Beams and lateral restraint−

Beams resist bending and shear; section geometry places material efficiently away from the neutral axis. Lateral restraint and load position influence lateral-torsional stability and deflection.

Study and application method

Draw load, support, shear-force/bending-moment tendency and compression flange; state whether restraint is present and what serviceability check is needed.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw load, support, shear-force/bending-moment tendency and compression flange; state whether restraint is present and what serviceability check is needed.

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Common mistake / precaution: Do not use a beam-table capacity without matching span, restraint, load case and section grade.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Trusses, Drawings, Fabrication and Erection

Roof truss form, purlins, joint details, loads, simple member force interpretation, working drawings, quantity awareness, fabrication/erection safety and inspection.

Approx. 14 hours **CO4** **Understand · Apply**

Learning goals

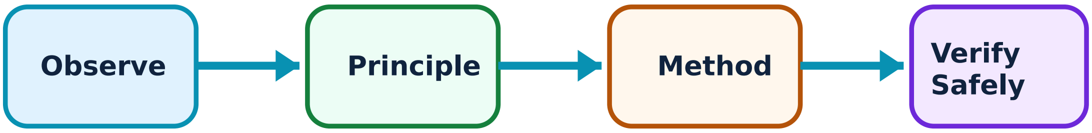
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Trusses, Drawings, Fabrication and Erection: identify the system, state its principle, follow the correct method and verify the result safely.

1. Roof trusses and load distribution−

A truss uses triangulated members to span efficiently. Roof loads pass through sheeting, purlins, truss members, supports and foundations; wind uplift and bracing require attention.

Study and application method

Sketch geometry, identify panels, purlin locations, support reactions and member force tendency; trace both gravity and wind load paths.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch geometry, identify panels, purlin locations, support reactions and member force tendency; trace both gravity and wind load paths.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Truss geometry and member forces must be analysed for actual load combinations—not estimated from a sketch.

2. Working drawings and quantities−

Steel drawings specify elevation, plan, member sizes/marks, connection details, weld/bolt notes and material information. A quantity take-off requires systematic member schedules and unit consistency.

Study and application method

Create a member schedule with mark, section, length, count, connection note and provisional mass source; cross-check against drawing views.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Create a member schedule with mark, section, length, count, connection note and provisional mass source; cross-check against drawing views.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Do not omit connection plates, fasteners, laps, wastage allowances or corrosion-protection items from a real estimate.

3. Fabrication, erection and inspection−

Fabrication and erection require sequencing, qualified work, temporary stability, lifting planning, inspection and protective treatment. Safety is integral to structural performance.

Study and application method

Prepare a simple method-statement outline: approved drawings, material traceability, fabrication checks, lifting/temporary bracing, inspection and handover records.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Never undertake lifting, hot work or structural alteration without competent supervision and approved procedures.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Steel, Sections, Loads and Design Basis

Use the concept flow to revise observation, principle, method and verification.

Module 2: Steel Connections

Use the concept flow to revise observation, principle, method and verification.

Module 3: Tension, Compression and Flexural Members

Use the concept flow to revise observation, principle, method and verification.

Module 4: Trusses, Drawings, Fabrication and Erection

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Identify standard steel section forms and trace load paths in a simple framed sketch.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare a bolted and welded connection drawing for load path, access and inspection needs.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare an annotated preliminary member-check worksheet using instructor-supplied current code data.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Read a simple roof-truss drawing and create a member/detail schedule with safety notes.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Steel, Sections, Loads and Design Basis

Explain Structural steel and section forms and state one correct method, application or precaution.–

Steel has high strength-to-weight ratio and ductility, but members must be selected for the force, connection, fabrication, corrosion and stability context. Common forms include angles, channels, I/H sections, tees, plates and hollow sections.

Answer framework: For a member, identify force type, likely failure/stability concern, feasible section family and the information that must be checked from a current standard/table.

Important point: Never choose a section from appearance or a catalogue size alone; verify grade, properties, slenderness, connection and load path.

Explain Loads, load paths and limit states and state one correct method, application or precaution.−

Dead, imposed, wind, seismic and other project loads travel through roof/floor systems, beams, columns, connections and foundations. Design considers safety and serviceability under specified combinations.

Answer framework: Sketch the structural system, draw force arrows, identify reactions and state the load combination/source that an authorised design would use.

Important point: Do not use isolated loads without applicable combinations, restraints and boundary conditions.

Explain Standards, drawings and design records and state one correct method, application or precaution.−

Design standards define material, load, detailing and verification requirements. Drawings communicate geometry, member marks, connections, tolerances and fabrication/erection information.

Answer framework: Read title block, grid, member schedule, section/detail callouts and notes; maintain units and a transparent calculation/drawing revision trail.

Important point: Only current project-approved codes and drawings may guide real construction.

Module 2 — Steel Connections

Explain Bolted connections and state one correct method, application or precaution.−

Bolted connections transfer forces through bearing, shear, friction or tension mechanisms depending on type and detailing. Layout, edge distances, spacing and hole details affect strength and constructability.

Answer framework: Draw the force path from member to connection to support; list bolt type/grade, load direction, likely failure modes and code checks required.

Important point: Do not invent bolt capacities or spacing; obtain them from current approved design provisions.

Explain Welded connections and state one correct method, application or precaution.−

Butt and fillet welds create continuous load transfer when designed, fabricated and inspected correctly. Weld size, length, orientation, access and quality control are critical.

Answer framework: Show weld symbol/location, resolve applied force into the weld group and identify required design/inspection information.

Important point: Welding safety, qualified procedures and inspection are essential; classroom calculations do not authorise field welding.

Explain Detailing and constructability and state one correct method, application or precaution.−

A good connection is not only strong: it is accessible for fabrication, erection, tightening/welding, inspection, drainage and maintenance. Eccentric load paths and abrupt geometry can create unintended stresses.

Answer framework: Review a connection drawing for load path, access, sequencing, edge clearances, corrosion protection and inspection points.

Important point: Never ignore erection stability or temporary bracing while focusing only on final connection strength.

Module 3 — Tension, Compression and Flexural Members

Explain Tension members and state one correct method, application or precaution.−

Tension members such as ties and truss elements can fail by yielding, rupture at net section or connection-related mechanisms. Section reduction at holes and connection geometry matter.

Answer framework: Identify force, connection layout and critical section; distinguish gross/net area concepts and list relevant current-code strength checks.

Important point: Do not assume the full section area remains effective where bolts, weld geometry or eccentricity alter force flow.

Explain Compression members and buckling and state one correct method, application or precaution.−

Compression members may fail by material yielding or instability. Effective length, end restraint, slenderness, imperfections and local buckling influence capacity.

Answer framework: Sketch end conditions and likely buckling axis, determine the data needed for effective length/slenderness and explain why bracing changes behaviour.

Important point: A member with sufficient material strength can still buckle; never neglect stability checks.

Explain Beams and lateral restraint and state one correct method, application or precaution.−

Beams resist bending and shear; section geometry places material efficiently away from the neutral axis. Lateral restraint and load position influence lateral-torsional stability and deflection.

Answer framework: Draw load, support, shear-force/bending-moment tendency and compression flange; state whether restraint is present and what serviceability check is needed.

Important point: Do not use a beam-table capacity without matching span, restraint, load case and section grade.

Module 4 — Trusses, Drawings, Fabrication and Erection

Explain Roof trusses and load distribution and state one correct method, application or precaution.−

A truss uses triangulated members to span efficiently. Roof loads pass through sheeting, purlins, truss members, supports and foundations; wind uplift and bracing require attention.

Answer framework: Sketch geometry, identify panels, purlin locations, support reactions and member force tendency; trace both gravity and wind load paths.

Important point: Truss geometry and member forces must be analysed for actual load combinations—not estimated from a sketch.

Explain Working drawings and quantities and state one correct method, application or precaution.−

Steel drawings specify elevation, plan, member sizes/marks, connection details, weld/bolt notes and material information. A quantity take-off requires systematic member schedules and unit consistency.

Answer framework: Create a member schedule with mark, section, length, count, connection note and provisional mass source; cross-check against drawing views.

Important point: Do not omit connection plates, fasteners, laps, wastage allowances or corrosion-protection items from a real estimate.

Explain Fabrication, erection and inspection and state one correct method, application or precaution.−

Fabrication and erection require sequencing, qualified work, temporary stability, lifting planning, inspection and protective treatment. Safety is integral to structural performance.

Answer framework: Prepare a simple method-statement outline: approved drawings, material traceability, fabrication checks, lifting/temporary bracing, inspection and handover records.

Important point: Never undertake lifting, hot work or structural alteration without competent supervision and approved procedures.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Steel, Sections, Loads and Design Basis.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Steel Connections.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Tension, Compression and Flexural Members.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.

7. State one key definition from Module 4: Trusses, Drawings, Fabrication and Erection.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Properties of structural steel, common rolled/built-up sections, section notation, loads, limit states, stability, drawings and code awareness..
2. Develop a complete answer or practical plan for: Bolted and welded connections, force transfer, detailing, eccentricity, fabrication, inspection and corrosion/fire considerations..
3. Develop a complete answer or practical plan for: Tension capacity, net/gross section concepts, column buckling/effective length/slenderness, restrained beams, lateral stability and serviceability..
4. Develop a complete answer or practical plan for: Roof truss form, purlins, joint details, loads, simple member force interpretation, working drawings, quantity awareness, fabrication/erection safety and inspection..

LAST-MILE REVISION

Quick Revision

Module 1: Steel, Sections, Loads and Design Basis

Properties of structural steel, common rolled/built-up sections, section notation, loads, limit states, stability, drawings and code awareness.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Steel Connections

Bolted and welded connections, force transfer, detailing, eccentricity, fabrication, inspection and corrosion/fire considerations.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Tension, Compression and Flexural Members

Tension capacity, net/gross section concepts, column buckling/effective length/slenderness, restrained beams, lateral stability and serviceability.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Trusses, Drawings, Fabrication and Erection

Roof truss form, purlins, joint details, loads, simple member force interpretation, working drawings, quantity awareness, fabrication/erection safety and inspection.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Structural steel and section forms	Steel has high strength-to-weight ratio and ductility, but members must be selected for the force, connection, fabrication, corrosion and stability context.
Loads, load paths and limit states	Dead, imposed, wind, seismic and other project loads travel through roof/floor systems, beams, columns, connections and foundations.
Standards, drawings and design records	Design standards define material, load, detailing and verification requirements.
Bolted connections	Bolted connections transfer forces through bearing, shear, friction or tension mechanisms depending on type and detailing.
Welded connections	Butt and fillet welds create continuous load transfer when designed, fabricated and inspected correctly.
Detailing and constructability	A good connection is not only strong: it is accessible for fabrication, erection, tightening/welding, inspection, drainage and maintenance.
Tension members	Tension members such as ties and truss elements can fail by yielding, rupture at net section or connection-related mechanisms.
Compression members and buckling	Compression members may fail by material yielding or instability.
Beams and lateral restraint	Beams resist bending and shear; section geometry places material efficiently away from the neutral axis.
Roof trusses and load distribution	A truss uses triangulated members to span efficiently.
Working drawings and quantities	Steel drawings specify elevation, plan, member sizes/marks, connection details, weld/bolt notes and material information.
Fabrication, erection and inspection	Fabrication and erection require sequencing, qualified work, temporary stability, lifting planning, inspection and protective treatment.

LEARNING RESOURCES

References

Recommended steel-structures references

1. S. K. Duggal, Limit State Design of Steel Structures.
2. N. Subramanian, Design of Steel Structures.
3. Current applicable BIS steel-structure standards and project specifications.

Approved public steel-structures sources

- https://hsbte.org.in/pdf/syllabus/CIVIL%20ENGG/6.pdf

- https://testbook.com/civil-engineering/design-of-steel-structures-notes

These sources support the study handbook while official Course 4011 content is unavailable; they do not replace official syllabus requirements or current code-based professional design.